

Nonnegative and positive factorizations

The maximum cp-rank in order six

Difficulty: challenging **Importance:** interesting to the community

Status: open; checked 2026-09-08

Area: size of nonnegative symmetric factorizations

Context and notation

A symmetric matrix A is **completely positive** if $A = BB^T$ for an entrywise nonnegative real matrix B with finitely many columns. Let $\mathcal{CP}_n$ be the cone of these matrices of order n . Its **cp-rank**, $\text{cpr}(A)$, is the minimum number of columns in such a factor, with $\text{cpr}(0) = 0$. Boundaries and interiors use the usual Euclidean topology on the vector space of real symmetric matrices.

Problem statement

Is every real completely positive matrix of order six a sum of at most nine nonnegative rank-one matrices? Equivalently, is

$$\max_{A \in \mathcal{CP}_6} \text{cpr}(A) = 9,$$

or, equivalently, does each $A \in \mathcal{CP}_6$ admit $A = BB^T$ with $B \in \mathbb{R}_{\geq 0}^{6 \times 9}$, allowing zero columns?

The lower bound nine is established. Order six is the unresolved case of the original Drew–Johnson–Loewy bound: order five satisfies it, whereas counterexamples exist in higher orders. It is included as the specific remaining case identified in the literature, rather than as one item in a dimension-by-dimension list.

A known reduction places a matrix attaining the order-six maximum on the boundary of $\mathcal{CP}_6$, with full ordinary rank and at least one zero entry. This reduction does not establish the nine-column bound.

References

Abraham Berman, Mirjam Dür, and Naomi Shaked-Monderer, *Open problems in the theory of completely positive and copositive matrices*, Electronic Journal of Linear Algebra **29** (2015), 46–58, §4.2, p. 53. Naomi Shaked-Monderer, *On the DJL conjecture for order 6, corrected arXiv v3*, 2017-06-01; journal version in Operators and Matrices **11** (2017), 71–88, Theorem 1.1 and Corollary 3.2.

Status evidence

Searches for “DJL conjecture 2026”, “DJL conjecture 2025”, “cp-rank order 6 solved”, and “maximum cp-rank 6 2026” found no order-six proof or counterexample. The corrected v3 retains the question. Roger Behling, Douglas Gonçalves, Hugo Lara, and Harry Oviedo, *A Projected Inexact Levenberg-Marquardt Method for the Completely Positive Matrix Factorization*, Journal of Optimization Theory and Applications **209** (2026), article 9, §5.6, reproduces higher-order counterexamples to the general DJL conjecture, not an order-six resolution. The most recent explicit open statement verified for this exact case is the corrected 2017 source.